

A Technical Introduction to Image captioning

Section 1

Grass pollen levels for Friday have increased from the moderate to high levels of yesterday with values of around 6 to 7 across most parts of the country. However, in Northern areas, pollen levels will be moderate with values of 4.

Section 2

Creative writing and computational humor Creative language generation by NLG has been hypothesized since the field's origins. A recent pioneer in the area is Phillip Parker, who has developed an arsenal of algorithms capable of automatically generating textbooks, crossword puzzles, poems and books on topics ranging from bookbinding to cataracts. The advent of large pretrained transformer-based language models such as GPT-3 has also enabled breakthroughs, with such models demonstrating recognizable ability for creating-writing tasks. A related area of NLG application is computational humor production. JAPE (Joke Analysis and Production Engine) is one of the earliest large, automated humor production systems that uses a hand-coded template-based approach to create punning riddles for children. HAHAAcronym creates humorous reinterpretations of any given acronym, as well as proposing new fitting acronyms given some keywords. Despite progresses, many challenges remain in producing automated creative and humorous content that rival human output. In an experiment for generating satirical headlines, outputs of their best BERT-based model were perceived as funny 9.4% of the time (while real headlines from The Onion were 38.4%) and a GPT-2 model fine-tuned on satirical headlines achieved 6.9%. It has been pointed out that two main issues with humor-generation systems are the lack of annotated data sets and the lack of formal evaluation methods, which could be applicable to other creative content generation. Some have argued relative to other applications, there has been a lack of attention to creative aspects of language production within NLG. NLG researchers stand to benefit from insights into what constitutes creative language production, as well as structural features of narrative that have the potential to improve NLG output even in data-to-text systems.

Section 3

Automatic report generation From a commercial perspective, the most successful NLG applications have been data-to-text systems which generate textual summaries of databases and data sets; these systems usually perform data analysis as well as text generation. Research has shown that textual summaries can be more

effective than graphs and other visuals for decision support, and that computer-generated texts can be superior (from the reader's perspective) to human-written texts. The first commercial data-to-text systems produced weather forecasts from weather data. The earliest such system to be deployed was FoG, which was used by Environment Canada to generate weather forecasts in French and English in the early 1990s. The success of FoG triggered other work, both research and commercial. Recent applications include the UK Met Office's text-enhanced forecast. Data-to-text systems have since been applied in a range of settings. Following the minor earthquake near Beverly Hills, California on March 17, 2014, The Los Angeles Times reported details about the time, location and strength of the quake within 3 minutes of the event. This report was automatically generated by a 'robo-journalist', which converted the incoming data into text via a preset template. Currently there is considerable commercial interest in using NLG to summarise financial and business data. Indeed, Gartner has said that NLG will become a standard feature of 90% of modern BI and analytics platforms. NLG is also being used commercially in automated journalism, chatbots, generating product descriptions for e-commerce sites, summarising medical records, and enhancing accessibility (for example by describing graphs and data sets to blind people). An example of an interactive use of NLG is the WYSIWYM framework. It stands for What you see is what you meant and allows users to see and manipulate the continuously rendered view (NLG output) of an underlying formal language document (NLG input), thereby editing the formal language without learning it. Looking ahead, the current progress in data-to-text generation paves the way for tailoring texts to specific audiences. For example, data from babies in neonatal care can be converted into text differently in a clinical setting, with different levels of technical detail and explanatory language, depending on intended recipient of the text (doctor, nurse, patient). The same idea can be applied in a sports setting, with different reports generated for fans of specific teams.

Section 4

Natural language generation (NLG) is a software process that produces natural language output. A widely cited survey of NLG methods describes NLG as "the subfield of artificial intelligence and computational linguistics that is concerned with the construction of computer systems that can produce understandable texts in English or other human languages from some underlying non-linguistic representation of information". While it is widely agreed that the output of any NLG process is text, there is some disagreement about whether the inputs of an NLG system need to be non-linguistic. Common applications of NLG methods include the production of various reports, for example weather and patient reports; image captions; and chatbots like ChatGPT. Automated NLG can be compared

to the process humans use when they turn ideas into writing or speech. Psycholinguists prefer the term language production for this process, which can also be described in mathematical terms, or modeled in a computer for psychological research. NLG systems can also be compared to translators of artificial computer languages, such as decompilers or transpilers, which also produce human-readable code generated from an intermediate representation. Human languages tend to be considerably more complex and allow for much more ambiguity and variety of expression than programming languages, which makes NLG more challenging. NLG may be viewed as complementary to natural language understanding (NLU): whereas in natural-language understanding, the system needs to disambiguate the input sentence to produce the machine representation language, in NLG the system needs to make decisions about how to put a representation into words. The practical considerations in building NLU vs. NLG systems are not symmetrical. NLU needs to deal with ambiguous or erroneous user input, whereas the ideas the system wants to express through NLG are generally known precisely. NLG needs to choose a specific, self-consistent textual representation from many potential representations, whereas NLU generally tries to produce a single, normalized representation of the idea expressed. NLG has existed since ELIZA was developed in the mid 1960s, but the methods were first used commercially in the 1990s. NLG techniques range from simple template-based systems like a mail merge that generates form letters, to systems that have a complex understanding of human grammar. NLG can also be accomplished by training a statistical model using machine learning, typically on a large corpus of human-written texts.